Klippel-Trenaunay Syndrome Complicated by Kasabach-Merritt Syndrome During Pregnancy

A case report and review of the literature

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Abstract

The co-occurrence of Klippel–Trenaunay syndrome and Kasabach-Merritt syndrome in pregnancy is extremely rare and represents a critical obstetric emergency with significant maternal and fetal risks, demanding early recognition and immediate multidisciplinary intervention.

We report the case of a 28-year-old pregnant woman with known Klippel–Trenaunay syndrome who developed Kasabach–Merritt phenomenon during pregnancy. The fetus suffered severe growth restriction and absent end-diastolic flow on umbilical artery Doppler. A cesarean section was performed at 35 weeks of gestation.

This case highlights the complexity of managing pregnancy in women with Klippel–Trenaunay syndrome complicated by Kasabach Merritt syndrome and emphasizes the importance of multidisciplinary surveillance and timely delivery.

Keywords: Klippel–Trenaunay syndrome; angioosteohypertrophy syndrome; Kasabach Merritt syndrome; fetal growth restriction; high-risk pregnancy

Introduction

Klippel-Trenaunay syndrome (KTS) is a rare congenital disorder affecting an estimated 2–5 per 100,000 individuals. Its features include a triad of capillary malformations, venous and lymphatic anomalies, and limb overgrowth. [1,2]

KTS was first described in 1900 as Naevus vasculosus osteohypertrophicus by two French physicians, Maurice Klippel and Paul Trenaunay. It was previously considered similar to the less common Parkes Weber syndrome, distinguished from KTS by the presence of arteriovenous malformations. [3,4]. KTS is currently classified within the PIK3CA-Related Overgrowth Spectrum (PROS), a group of overgrowth syndromes with overlapping phenotypic features that share mutations in the PIK3CA gene. KTS, also referred to as capillary-lymphatic-venous malformation, is characterized by capillary malformations, soft tissue or bony hypertrophy, and varicose veins or venous malformations. The presence of at least two of these features suggests the diagnosis, which is supported by imaging. Magnetic resonance imaging, Color Doppler studies, and pelviabdominal ultrasounds may all be used to visualize vascular malformations associated with KTS.[5] KTS hallmark features predominantly affect one lower limb (85%) causing a discrepancy in limb length, although bilateral involvement may occur in a minority of cases (12.5%). However, it is a generalized disorder and can involve multiple body regions including the pelvis, abdomen, head and neck, back, pelvis, and urinary bladder. Capillary malformations are the most common presentation (98%), followed by varicosities or venous malformations (72%), and limb hypertrophy (67%) [6].

Pregnancy poses considerable challenges in patients with KTS, as hormonal and hemodynamic changes exacerbate underlying vascular abnormalities increasing the risk of thromboembolism, hemorrhage, and coagulopathy. The emergence of coagulopathy, in particular, may indicate the development of Kasabach-Merritt syndrome (KMS). [7]

Kasabach-Merritt syndrome (KMS), first described in 1940, is a rare and potentially fatal condition defined by the triad of thrombocytopenia, microangiopathic hemolytic anemia, and consumptive coagulopathy occurring in association with vascular abnormalities [8]. In the setting of KTS, pregnancy and labor may precipitate or exacerbate KMS; the concurrent occurrence of both conditions during the peripartum period constitutes a critical obstetric emergency carrying a reported mortality rate of 30–40% [9].

To date, only three cases of KTS complicated by KMS in pregnancy have been documented in the literature [10–12]. We report a fourth case, describing the clinical course and successful multidisciplinary management of a pregnant woman with KTS complicated by fetal growth restriction and peripartum acute KMS. Through this case, we aim to illuminate the clinical challenges inherent to this rare complication, outline effective management strategies, and emphasize the critical role of early recognition and interdisciplinary collaboration in achieving favorable maternal and neonatal outcomes.

Case Presentation

A 28-year-old multigravida (P2+1, both prior deliveries by cesarean section, one living child) presented at 30+1 weeks' gestation with clinically significant hematuria and was admitted on December 16th to Ain Shams University Hospital, Egypt. The clinical timeline is detailed in Table 1.

Her background was notable for Klippel-Trenaunay syndrome (KTS), diagnosed at age two, with involvement of both lower limbs, the back, pelvis, and urinary bladder. Her surgical history included varicose vein surgery seven years prior and orthopedic fixation with plates and screws approximately 18 months before this admission.

Her obstetric history was significant. Her first pregnancy, seven years prior, ended in neonatal death. Her second pregnancy was complicated by severe preeclampsia, postpartum wound hematoma, and sepsis necessitating ICU admission. She also had a history of spontaneous miscarriage not managed surgically. In the current pregnancy, she received two doses of antenatal corticosteroids on December 14th and 15th.

On admission, vital signs were stable: blood pressure 120/80 mmHg, temperature 36.2°C, and heart rate 68 bpm. Physical examination revealed port-wine stains over the back and gluteal region, with discrepancy in lower limb length bilaterally.

A complete blood count (CBC) demonstrated microcytic hypochromic anemia (hemoglobin 7.0 g/dL, hematocrit 23.0%) and thrombocytopenia (128,000/µL). She received six units of packed red blood cells (pRBCs) to correct her declining hemoglobin. Blood typing identified B-positive blood group, and viral serology was negative for hepatitis B, hepatitis C, and HIV.

Obstetric ultrasonography revealed an estimated fetal weight below the 10th percentile, consistent with intrauterine growth restriction (IUGR), and absent end-diastolic flow on umbilical artery Doppler. Urological consultation was obtained for management of severe hematuria, with initial measures comprising intravenous tranexamic acid and insertion of a three-way indwelling silicone urinary catheter maintained for one week.

On December 18th, pelviabdominal MRI demonstrated a viable single intrauterine pregnancy in cephalic presentation with unremarkable placental location and attachment. Compression of the left ovarian and left common iliac veins by the gravid uterus was identified, with associated venous dilatation and slowing of flow. Extensive dilated, tortuous tubular lesions consistent with varicosities were observed throughout the pelvic, perivesical, perineal, and gluteal regions — more prominent on the left — exhibiting a classic "bag of worms" appearance. Similar lesions were identified in the bilateral breast subcutaneous tissue. The urinary bladder contained blood products. The liver was mildly enlarged with vascular compression, while the gallbladder, spleen, and kidneys were unremarkable. Minimal bilateral pleural effusion was also noted.

On December 21st, pelviabdominal ultrasonography, color Doppler of bilateral lower limb veins, and duplex arterial studies were performed. Ultrasonography demonstrated hepatosplenomegaly, soft tissue lesions within the urinary bladder, and bilateral refluxing parametrial venous channels, with a single viable intrauterine fetus confirmed. Lower limb vascular studies revealed a normal venous and arterial system with no evidence of deep vein thrombosis.

Serial CBCs over the four days following transfusion showed progressive improvement in hemoglobin, reaching 9.4 g/dL on December 22nd, while platelet count remained persistently low, ranging between 82,000 and 104,000/µL. Iron studies revealed elevated unsaturated iron-binding capacity (UIBC) and reduced transferrin saturation. Lactate dehydrogenase (LDH) was elevated at 454 IU/L. Urinalysis on the same day demonstrated hematuria (2+), pyuria, and mild proteinuria (1+); microscopic examination showed >100 red blood cells per high-power field (RBCs/HPF), 40–45 pus cells/HPF, increased epithelial cells (2+), and uric acid crystals (2+). Coagulation studies on December 27th and 28th revealed mildly elevated prothrombin time (PT), while serum creatinine remained low.

Contrast-enhanced pelvic MRI on January 1st demonstrated extensive venous malformations involving the cervix, vagina, anal canal, perineum, and bilateral labia, with additional lesions in the gluteal region, thighs, anterior abdominal wall, and intravesically.

The patient was monitored daily with regular assessment of vital parameters, fetal kick counts, vaginal bleeding, uterine contractions, and signs of membrane rupture. She received one unit of pRBCs on January 2nd and a further unit on January 5th.

At 35 weeks' gestation, a multidisciplinary team comprising obstetrics, hematology, anesthesia, and neonatology convened and recommended delivery by emergency cesarean section in view of deteriorating fetal Doppler parameters and cumulative maternal risks. On January 18th, cesarean delivery was performed under general endotracheal anesthesia via a vertical midline abdominal incision. Intraoperatively, significant varicosities were encountered over the anterior abdominal wall, lower uterine segment, and the uterovesical interface. A lower uterine segment incision was made for delivery of the fetus and placenta; the uterine wall was subsequently closed in two layers. The placenta was delivered complete, the perineum was intact, and hemostasis was secured. No intraperitoneal drains were placed; a subcutaneous drain was inserted and maintained for seven days. Skin closure was achieved with interrupted mattress sutures.

A female neonate weighing 1,900 g was delivered with Apgar scores of 7 and 9 at one and five minutes, respectively. Given the low birth weight, routine vaccination was deferred, and neither observation nor respiratory support was required.

Postoperatively, the patient was admitted to the ICU following identification of hemoperitoneum. Her pharmacological regimen included low-molecular-weight heparin (LMWH), tranexamic acid, ceftriaxone, pantoprazole, intravenous paracetamol, and a charcoal-based herbal preparation.

On postoperative day one, mild hematuria persisted and laboratory investigations revealed declining hemoglobin and platelet counts alongside markedly elevated LDH, alkaline phosphatase, creatine kinase, and bilirubin. LMWH was withheld, and the patient was commenced on pRBCs and fresh frozen plasma (FFP) in a 1:1 ratio, receiving a total of ten units of each over seven days. Additional laboratory results are presented in @tbl-2.

On postoperative day two, hemoglobin and platelet count deteriorated rapidly, falling from 9.6 g/dL and 218,000/µL to 7.6 g/dL and 104,000/µL, respectively, necessitating ongoing blood product replacement.

On postoperative day three, transvaginal ultrasonography (TVUS) revealed marked hemoperitoneum with a localized lower uterine segment hematoma measuring approximately 12 cm in maximum diameter. Conservative management was elected by the attending consultant and departmental head. The pharmacological regimen was revised to include LMWH, tranexamic acid, calcium, dexamethasone, magnesium sulfate, cefepime, omeprazole, clindamycin, and intravenous paracetamol.

On postoperative day four, coagulation studies revealed hypofibrinogenemia (fibrinogen 1.4 g/L) and markedly elevated D-dimer (35.2 µg/mL FEU), findings consistent with a diagnosis of Kasabach-Merritt syndrome (KMS).

Urinalysis on day six demonstrated persistent hematuria (2+), mild pyuria, and mild proteinuria (1+), with microscopic examination showing >100 RBCs/HPF, 13–15 pus cells/HPF, and increased epithelial cells (1+). By day seven, hemoglobin had stabilized at 10.1 g/dL and remained stable over the subsequent three days, though thrombocytopenia persisted at 64,000/µL.

Following clinical stabilization, the patient was discharged on postoperative day ten. At her outpatient review on day nineteen, she was in satisfactory general condition with a soft, lax abdomen; skin sutures were removed and the abdominal wound demonstrated optimal healing.

Table 1: Patient timeline

Date	Timeline
30 +1 weeks' gestation (Dec 16, 2025)	Admission of a case of KTS, presenting with hematuria; anemia and thrombocytopenia; fetus <10th percentile, received tranexamic acid and 6 units PRBCs.
30+1 to 35 weeks' gestation (Dec 18, 2025 – Jan 17, 2026)	MRI: extensive venous malformations (pelvis, thigh, abdomen). Close maternal/fetal monitoring. Intermittent transfusions slight hemoglobin improvement; fluctuating platelets; prolonged PT.
35 weeks (Jan 18, 2026)	Emergency cesarean section for fetal compromise → female neonate (1900 g). Mother admitted to ICU, hemoperitoneum managed conservatively.
Postoperative course	**Day 1:** Transfusion support (total 10 units PRBC + FFP over 7 days) **Day 2:** Hb 9.6→7.6 g/dL; platelets $218\times10^3 \rightarrow 104\times10^3/\mu L$ **Day 3:** Conservative management, LMWH resumed, tranexamic acid initiated **Day 4:** Hypofibrinogenemia (1.4 g/L), elevated D-dimer (35.2 µg/mL) consistent with Kasabach–Merritt syndrome **Day 7:** Hb stabilized (10.1 g/dL); persistent thrombocytopenia ($64\times10^3/\mu L$) **Day 10:** Discharged in stable condition **Day 19:** Outpatient follow-up: fair general condition; sutures removed

Table 2: Laboratory results throughout the clinical course.

Parameters/Timing	December 16 (Day of admission)	December 22	December 28	POD 1	POD 2	POD 4	POD 6	POD 7	Reference range
Hb (g/dL)	7.0	7.6	10.2	9.6	7.6	8.6	10.0	10.0	12.0-15.0
HCT (%)	23.0	23.5	33.1	20.0	21.8	26.8	30.8	-	36-46
Platelets (/µL)	128,000	97,000	129,000	218,000	104,000	75,000	64,000	64,000	150,000–410,000

Parameters/Timing	December 16 (Day of admission)	December 22	December 28	POD 1	POD 2	POD 4	POD 6	POD 7	Reference range
PT (seconds)	-	-	12.4	40.70	-	-	17.7	-	11.7–15.3
Total bilirubin (mg/dL)	-	-	-	2.8	1.8	-	1.4	-	1.8-3.5
ALP (IU/L)	-	-	-	135	115	-	108	-	34–104
LDH (IU/L)	-	454	-	424	357	-	449	-	140–271

POD: Post operative day, Hb: Hemoglobin, HTC: Hematocrit, PT: Prothrombin time, aPTT: Activated partial thromboplastin time, ALP: Alkaline phosphatase, LDH : Lactate dehydrogenase, BUN: Blood Urea Nitrogen.

Review of literature

A systematic search of MEDLINE, Embase, and Scopus identified only three reported cases of KTS complicated by peripartum KMS. KTS was diagnosed prior to pregnancy in two cases [10,11], whereas in the third it had been misdiagnosed as acromegaly and was not recognized until the final trimester [12]. All cases presented with classic KTS features — multiple varicosities, limb hypertrophy, and hemangiomas — with one additionally demonstrating port-wine stains [12]. Despite treatment, postpartum laboratory parameters failed to normalize, indicating refractory coagulopathy consistent with KMS.

All three patients had marked vulvar varicosities necessitating cesarean delivery, with intra-operative or postoperative hemorrhage requiring multiple transfusions; two required massive transfusion protocols [10,12]. Management followed a multidisciplinary approach in all cases, incorporating anticoagulation and blood product support. Two cases received corticosteroids and antifibrinolytics [10,11]. One patient developed life-threatening hemorrhage requiring postpartum hysterectomy, complicated further by DIC, sepsis, peritonitis, and hemoperitoneum necessitating emergency laparotomy [12]. All patients required prolonged postoperative hospitalization. None of the neonates demonstrated clinical or laboratory evidence of KTS.

Detailed summary of the three cases is shown in Table 3

Table 3: Literature review

Author (Year)	Pregnancy	**Key findings**	**KMS**	**Management**	**Outcome**
Neubert (1995) [10]	22 years-old primigravida, 37 weeks' gestation	known KTS, hemangiomas, limb hypertrophy, massive vulvovaginal varicosities	Post-op KMS: thrombocytopenia, decreased fibrinogen, prolonged PT/PTT, increased fibrin split products	CS at 39 wks; hematoma evacuation; ICU; transfusion; heparin + aminocaproic acid	Discharged day 20; healthy neonate
Kanmani (2021) [11]	24 years-old, primigravida, 29 weeks' gestation	known KTS, bleeding gums, Limb hypertrophy, varicosities (vulvar, lower limb)	Antenatal KMS: severe thrombocytopenia (limited laboratory investigation reported)	Steroids + platelet transfusion; CS at 35 wks; post-op anticoagulation and corticosteroids	Discharged day 20; healthy neonate
Zhang (2019) [12]	26 years old. Postpartum.	KTS diagnosed in the third trimester of pregnancy. Admitted on day 10 postpartum with severe postpartum hemorrahge, hemodynamic instability, ecchymoses, massive limb hypertrophy, varicosities	Post-op KMS: thrombocytopenia, ↓ fibrinogen, ↑ PT/APTT, increased fibrin split products, increased D-dimer	CS at 39 wks; hysterectomy; embolization; laparotomy; anticoagulation	Discharged day 42; healthy neonate

Discussion

The occurrence of Klippel-Trenaunay syndrome complicated by Kasabach-Merritt syndrome during pregnancy is exceedingly rare, with only three such cases reported in the literature prior to this report [10–12]. The present case thus represents the fourth documented instance of this life-threatening combination, and the first to describe its successful management in the context of concomitant fetal growth restriction and peripartum acute KMS.

Diagnosis and baseline complexity

Our patient presented with an unusually extensive KTS phenotype, with disease involvement encompassing both lower limbs, the back, pelvis, urinary bladder, and breast tissue — a distribution reflecting the multisystem vascular dysplasia characteristic of KTS. Consistent with previously reported cases [10–12], she exhibited the hallmark triad of varicosities, limb hypertrophy, and hemangiomas, with port-wine stains additionally present over the gluteal region. Her obstetric history further compounded the clinical complexity: two prior cesarean sections, a neonatal death, a previous pregnancy complicated by severe preeclampsia and ICU admission, and a spontaneous miscarriage — underscoring the cumulative maternal morbidity associated with KTS across successive pregnancies.

Clinically significant hematuria at 30+1 weeks' gestation was the sentinel presentation leading to admission, a manifestation attributable to intravesical venous malformations confirmed on both MRI and contrast-enhanced pelvic imaging. Notably, pelviabdominal MRI revealed extensive pelvic varicosities with a characteristic "bag of worms" appearance, compression of the left ovarian and common iliac veins by the gravid uterus, hepatosplenomegaly, and bilateral pleural effusion — findings collectively reflecting the hemodynamic burden imposed by the gravid state on an already compromised vascular system. This aligns with prior observations that pregnancy-associated hormonal and hemodynamic changes exacerbate the vascular pathology inherent to KTS, amplifying risks of thromboembolism and hemorrhage [[13]].

Antenatal management

Initial antenatal management was guided by two principal concerns: controlling hematuria and optimizing fetal and maternal status ahead of inevitable preterm delivery. Hematuria was managed conservatively with intravenous tranexamic acid and continuous bladder irrigation via a three-way urinary catheter, achieving partial resolution without surgical intervention. Serial CBCs demonstrated persistent thrombocytopenia throughout the antenatal period, with platelet counts ranging between 82,000 and 128,000/µL despite transfusion support, accompanied by elevated LDH — early biochemical signals of evolving consumptive coagulopathy. These antenatal laboratory aberrations, in retrospect, likely reflected subclinical KMS activity preceding its overt peripartum manifestation.

Fetal surveillance revealed IUGR with absent end-diastolic umbilical artery flow, indicating significant uteroplacental insufficiency. The administration of two doses of antenatal corticosteroids at 30 weeks represented an important adjunct in preparing for anticipated preterm delivery. The decision to deliver at 35 weeks was reached through multidisciplinary consensus, balancing the risk of ongoing fetal compromise against the substantial maternal surgical risks posed by the extensive pelvic varicosities. This mirrors the management philosophy reported in all previously published cases, where cesarean delivery was ultimately necessitated by vulval or pelvic varicosities precluding safe vaginal birth [10–12].

Intraoperative findings and delivery

Cesarean section was performed under general endotracheal anesthesia, with the choice of a vertical midline incision allowing adequate visualization and surgical access in the setting of extensive anterior abdominal wall and lower uterine segment varicosities. Intraoperatively, significant varicosities were encountered at the uterovesical interface — a finding consistent with the MRI-demonstrated pelvic venous malformations and a recognized source of hemorrhagic risk in KTS patients undergoing cesarean delivery. Hemostasis was secured without recourse to uterine artery ligation, balloon tamponade, or hysterectomy, distinguishing the intraoperative course of this case favorably from that of a previously published case report [12], in which life-threatening hemorrhage required postpartum hysterectomy along with emergency laparotomy for DIC, sepsis, peritonitis, and hemoperitoneum.

A female neonate weighing 1,900 g was delivered with reassuring Apgar scores of 7 and 9 at one and five minutes, respectively. Consistent with all three previously reported cases [10–12], the neonate demonstrated no clinical features of KTS, supporting the current understanding that KTS does not follow a straightforward vertical transmission pattern.

Peripartum KMS and postoperative course

The postoperative course was complicated by hemoperitoneum requiring ICU admission, with a rapidly deteriorating hemoglobin and platelet count on days one and two, accompanied by markedly elevated LDH, alkaline phosphatase, creatine kinase, and bilirubin. On postoperative day three, TVUS confirmed a 12-cm lower uterine segment hematoma. The definitive diagnosis of KMS was established on day four, supported by hypofibrinogenemia (1.4 g/L) and markedly elevated D-dimer (35.2 µg/mL FEU) — laboratory findings fulfilling the diagnostic criteria of consumptive coagulopathy in the context of underlying vascular abnormalities [[8]].

Management of peripartum KMS in this case required careful navigation of the competing hemostatic risks inherent to the syndrome: thrombocytopenia and consumptive coagulopathy predispose to hemorrhage, while the vascular malformations underlying KMS simultaneously confer thrombotic risk. LMWH was initially withheld in response to hemorrhagic deterioration, then judiciously resumed alongside tranexamic acid once the coagulopathic picture became clearer. This anticoagulation-antifibrinolytic combination reflects the dual-pathway pathophysiology of KMS and is consistent with the approach described in two of the three previously reported cases [10,11]. Corticosteroid therapy, in the form of dexamethasone, was incorporated given its proposed role in reducing vascular lesion activity and platelet trapping. The patient ultimately received ten units each of pRBCs and FFP in a 1:1 ratio over seven days — constituting a massive transfusion — alongside calcium, magnesium sulfate, and broad-spectrum antibiotics for infective prophylaxis.

Hemoglobin stabilized at 10.1 g/dL by day seven, though thrombocytopenia persisted at 64,000/µL at discharge — a pattern consistent with the refractory coagulopathy observed in all previously reported cases, where postpartum laboratory parameters failed to normalize despite active treatment [10–12]. The patient was discharged on day ten and demonstrated satisfactory wound healing at her day-nineteen outpatient review, representing a favorable outcome relative to the severe postoperative morbidity — including sepsis, peritonitis, DIC, and hysterectomy — reported in the most complicated previously published case [[12]].

Multidisciplinary care and broader implications

The successful outcome in this case was underpinned by early multidisciplinary involvement across obstetrics, hematology, anesthesia, neonatology, urology, and radiology — a consensus echoed across all published cases and reflective of the inherent complexity of managing concurrent KTS and KMS in pregnancy. Early recognition of KMS, even in its subclinical antenatal form, appears critical: the persistent antenatal thrombocytopenia and elevated LDH in our patient, though not initially diagnostic, warranted heightened vigilance and likely informed the decision to deliver at 35 rather than a later gestational age.

This case further highlights the importance of preconception counseling in women with KTS. The published cross-sectional data [13] documenting significantly elevated risks of DVT, pulmonary embolism, and severe postpartum hemorrhage in KTS patients provides epidemiological context for the individual complexity illustrated here. Our patient's cumulative obstetric history reflects the substantial reproductive morbidity that KTS may confer across successive pregnancies and underscores the need for individualized, specialist-led preconception assessment in this population.

Conclusion

Pregnancy in women with Klippel–Trenaunay syndrome complicated by Kasabach–Merritt syndrome is extremely rare and poses significant diagnostic and therapeutic challenges. Early suspicion and diagnosis with intensive antenatal surveillance and multidisciplinary management are crucial to navigate the complex balance between thrombosis and bleeding and to optimize maternal and perinatal outcomes in such high-risk pregnancy and childbirth.

Abbreviation

Klippel-Trenaunay syndrome (KTS)

Kasabach Meritt Syndrome (KMS)

Magnetic resonance imaging (MRI)

Ultrasonography (US)

Transvaginal ultrasound (TVUS)

Intensive care unit (ICU)

Low-molecular-weight heparin (LMWH)

Lactate dehydrogenase (LDH)

Blood Urea Nitrogen (BUN)

Prothrombin time (PT)

Partial Thromboplastin Time (PTT)

Complete blood picture (CBC)

Disseminated Intravascular Coagulation (DIC)

Multidisciplinary team (MDT)

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

A written consent for publication has been obtained from the patient.

Availability of data and materials

The data and materials supporting the conclusions of this article are available as a supplementary materials.

Competing interests

The authors declare that they have no competing interests.

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Authors' contributions

SIA, ME, AFN contributed to writing the first draft of the manuscript. All the authors revised the manuscript critically for important intellectual content. All authors approved the final version of the manuscript.

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